DATA 202 — Quiz 4 Retake: Week 5

ANSWER KEY — Version A

Question 1 (SLO 05A) — Retake from Quiz 3

student	loans_per_month	hours_per_week	closest
S31	2	6	A
S32	6	3	B
S33	3	4	A
S34	5	1	B
S35	1	5	A
S36	7	2	B

New centroid A = (2, 5), new centroid B = (6, 2) – the mean loans_per_month and mean hours_per_week of each centroid's students.

Assign: each student goes to the centroid at the smallest straight-line distance. Update: each centroid moves to the average of its students.

Question 2 (SLO 05B) — Retake from Quiz 3

k = 4. Inertia: the elbow is at 4 – it falls steeply from k = 1 to 4 (1040 → 658 → 379 → 185), then only slowly (164, 151, 139, 126). Silhouette: highest at k = 4 (0.59). Both agree. (Choosing k = 8 just because its inertia is lowest is wrong: inertia always drops as k grows.)

Question 3 (SLO 05C) — Retake from Quiz 3

```
PCA().fit(X_scaled).explained_variance_ratio_.round(3)
# [0.456  0.306  0.195  0.043]
```

2 components: 0.456 + 0.306 = 0.762, so they keep **76.2%** (PC1 alone is only 45.6%).